DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology

Rachit Agarwal
School of Computer Science and Engineering
VIT Chennai, Chennai, India
rachit.agarwal2022@vitstudent.ac.in

Patel Kavya Vipulkumar
School of Computer Science and Engineering
VIT Chennai, Chennai, India
kavya.patel2022@vitstudent.ac.in

Aarav Prawin
School of Computer Science and Engineering
VIT Chennai, Chennai, India
aarav.prawin2022@vitstudent.ac.in

Dr. Sindhu Ravindran
School of Computer Science and Engineering
VIT Chennai, Chennai, India
sindhu.ravindran@vit.ac.in

Abstract—Skin-lesion classifiers rarely fail for lack of accuracy on clean benchmarks. They fail because they answer every case with the same apparent confidence: a blurred photograph, missing history, or a photograph and a history that point to different diagnoses all produce a label. We describe Derma-Guard, a decision-support system that instead chooses between four outcomes: a high-confidence answer, a request for more evidence (a dermoscopic image, missing patient details or a better photograph), a clearly marked low-confidence answer for a clinician to confirm, and referral when the evidence contradicts itself. Six specialists (evidential heads for the photograph, dermoscopy and metadata, an external legacy model, a nearest-neighbour vote and a class-balanced embedding classifier) are profiled case by case by an evidence manager, and a learned gate decides how much each should count; the gate was chosen from seven combination rules under five-fold cross-fitting. A calibrated, cost-aware policy with a Learn-then-Test bound on missed cancers makes the final choice. On 787 held-out MILK10k lesions, 89.1% of patients receive an answer and 57.1% a high-confidence one that is 77.1% accurate over eleven diagnoses and misses 3.1% of cancers, against 4.9% for a learned rejector ($p = 0.011$) and 3.8% for confidence thresholding at the same coverage. Across four fusion back-ends the policy's miss rate stays between 3.0% and 3.5%, while thresholding ranges from 3.5% to 9.3%: abstention by confidence is only as safe as the probabilities beneath it. Controlled image damage shows that evidential uncertainty misses blur, darkness, noise and occlusion; explicit checks remove every confident answer on damaged input. Without retraining, the system stays within its safety target on PAD-UFES-20 and Derm7pt, while its fine-grained diagnosis does not transfer. Re-evaluating our earlier prototype, whose reported accuracies of 84–90% were not reproducible, places its models at 57–75% on unseen patients.

Index Terms—dermatology, multimodal fusion, evidential deep learning, selective prediction, conformal prediction, risk control, evidence acquisition, clinical decision support

I. Introduction

A patient who photographs a mole and asks an app whether it is dangerous gets an answer almost every time. The answer comes with a number, often above 90%, and that number looks the same whether the photograph was sharp or blurred, whether the patient's age and the lesion's location agree with the image, and whether the lesion resembles anything in the training data. Deep networks reached dermatologist-level accuracy on curated images years ago [1], so accuracy on benchmarks is no longer the main obstacle to safe use. Behaviour is.

A careful dermatologist behaves differently. When the evidence is clear they give a diagnosis. When it is not, they look more closely with a dermatoscope, give a provisional opinion and ask for confirmation, or refer the patient when the history and the lesion tell different stories. We wanted to know whether a model can be built to behave this way, and how much it costs.

This paper is also the second attempt. The first version of Derma-Guard already had the right intuitions (a panel of specialist models, a sceptical component that questions their evidence, and trust that depends on the suspected diagnosis), but its weights were typed in by hand. When we audited it, its reported accuracies did not survive: the split was made by photograph rather than by patient, and its evaluation code adjusted predictions toward preset accuracy targets (Section V-E). We kept the ideas, replaced every hand-set number with one fitted on validation data, and made every result a script that can be re-run.

Contributions. (i) A six-specialist team whose per-case weights are learned from all of the evidence manager's

signals, chosen from seven rules by cross-fitted validation, with an exhaustive subset study and an oracle ceiling. (ii) A calibrated four-outcome policy that can request three kinds of missing evidence and certifies the missed-cancer rate of its confident answers. (iii) The finding that confidence-based abstention is only as safe as the fusion beneath it, whereas the calibrated policy is stable across fusion back-ends. (iv) A controlled degradation benchmark, two external tests without retraining, and an honest re-evaluation of the earlier prototype's seven models. (v) A working care platform, with every system variant available side by side in its interface.

II. Related Work

Skin-lesion classification. After Esteva et al. [1], the ISIC challenges [2] and datasets such as HAM10000 [3] made dermoscopic classification a standard benchmark, usually on ImageNet-pretrained residual networks [4]; dermatology foundation models such as PanDerm [5] are the current frontier. Multimodal work combines images with metadata through attention-like gating [6] or uses paired clinical and dermoscopic images [7]. PAD-UFES-20 [8] and MILK10k [9] provide smartphone photographs with metadata and paired clinical–dermoscopic images, respectively, and multi-task models can recommend when dermoscopy is needed [10]. None of these decide whether a prediction should be issued at all.

Uncertainty and evidential fusion. Evidential deep learning [11] places a Dirichlet over class probabilities and yields a per-case ignorance mass grounded in subjective logic [12] and Dempster–Shafer theory [13], [14]. Trusted multi-view classification [15] combines per-view opinions with Dempster's rule so that uninformative views fall silent. Temperature scaling [16] and isotonic regression [17] calibrate confidences, and Mahalanobis distances [18] with shrinkage covariance [19] detect unusual inputs.

Abstention, deferral and risk control. Chow's rule [20], selective classification [21] and learning to defer [22] decide when not to answer; AUGRC [23] evaluates them. Conformal prediction [24]–[27], including class-conditional variants [28], gives distribution-free set guarantees, and Learn-then-Test [29] and conformal risk control [30] extend them to general risks. Our contribution is their integration in one decision with learned multi-specialist evidence and measured evidence acquisition, evaluated against simple baselines on the same probabilities.

III. Data and Evaluation Protocol

MILK10k [9] has 5,240 lesions, each with one clinical close-up and one dermoscopic image, eleven diagnoses (95.7% confirmed by histopathology; 71.7% malignant), age, sex, body site, a skin-tone class and seven automatically computed concept scores. Without patient identifiers, we split by lesion into 3,667 training, 786 validation and 787 test lesions, asserting that no lesion appears twice. Validation fits everything that is fitted after the network: fusion rules, reliabilities, calibration, conformal quantiles, the risk threshold, the acquisition estimator and every choice between methods. The test split was scored once. The data are heavily imbalanced: basal cell carcinoma makes up 48% of lesions, while four classes have fewer than ten test lesions each, which is why we report macro-F1 and balanced accuracy next to accuracy throughout.

External data. PAD-UFES-20 [8] (2,298 smartphone photographs, six classes, no dermoscopy) and Derm7pt [7] (1,003 lesions with both images after mapping its diagnoses onto MILK10k classes) are used only for testing, without retraining or recalibration.

Metrics. For a system that may abstain we report coverage (share of high-confidence answers), accuracy among them and the missed-cancer rate (share of answered malignant lesions called benign), always comparing systems at equal coverage; macro-F1 and balanced accuracy for the rare classes; malignant-vs-benign AUROC; AUGRC [23]; acquisition efficiency $\mathrm{AE} = (G_\pi - G_{\mathrm{rand}})/(G_{\mathrm{oracle}} - G_{\mathrm{rand}})$ for the choice of whom to send for dermoscopy; and 95% bootstrap intervals with paired tests (2,000 resamples).

IV. Method

Fig. 1 shows the pipeline. A case moves left to right: specialists give opinions, the evidence manager profiles them, a fusion rule turns the profile into weights, and the policy decides.

A. Specialists

S1–S3: evidential heads. An ImageNet ResNet50 [4] shared by both image types and a residual metadata network produce three 256-dimensional embeddings, each feeding an evidential head with non-negative evidence e_m [11]. With $\alpha_m = e_m + 1$ and $S_m = \sum_k \alpha_{m,k}$, belief and ignorance are $b_{m,k} = e_{m,k}/S_m$ and $u_m = K/S_m$; a head contributes its pignistic distribution $p_m = b_m + u_m/K$. The heads are also combined by Dempster's rule [15],

$$C = \sum_{i \neq j} b_{1,i} b_{2,j}, \quad b_k = \frac{b_{1,k} b_{2,k} + b_{1,k} u_2 + b_{2,k} u_1}{1 - C}, \quad u = \frac{u_1 u_2}{1 - C}, \tag{1}$$

which leaves a missing modality ($u = 1$) exactly without effect and separates contradiction (C) from shared ignorance (u). The network was trained on a Kaggle GPU for 25 epochs with modality dropout 0.25, so that it learns to say "I don't know" for an absent input.

S4: legacy model. The earlier prototype's image model, trained on PAD-UFES-20, is applied unchanged; its six classes are mapped to MILK10k and mixed with 10% uniform mass so unsupported classes are unknown rather than impossible.

S5, S6: case-based specialists. S5 is a vote of the 15 most similar training lesions in embedding space, weighted by $\exp(5\,\mathrm{sim})$ [31]. S6 is a multinomial logistic regression over all training embeddings with class weights inversely proportional to frequency, so that rare diagnoses are not drowned out; its regularisation was chosen on validation.

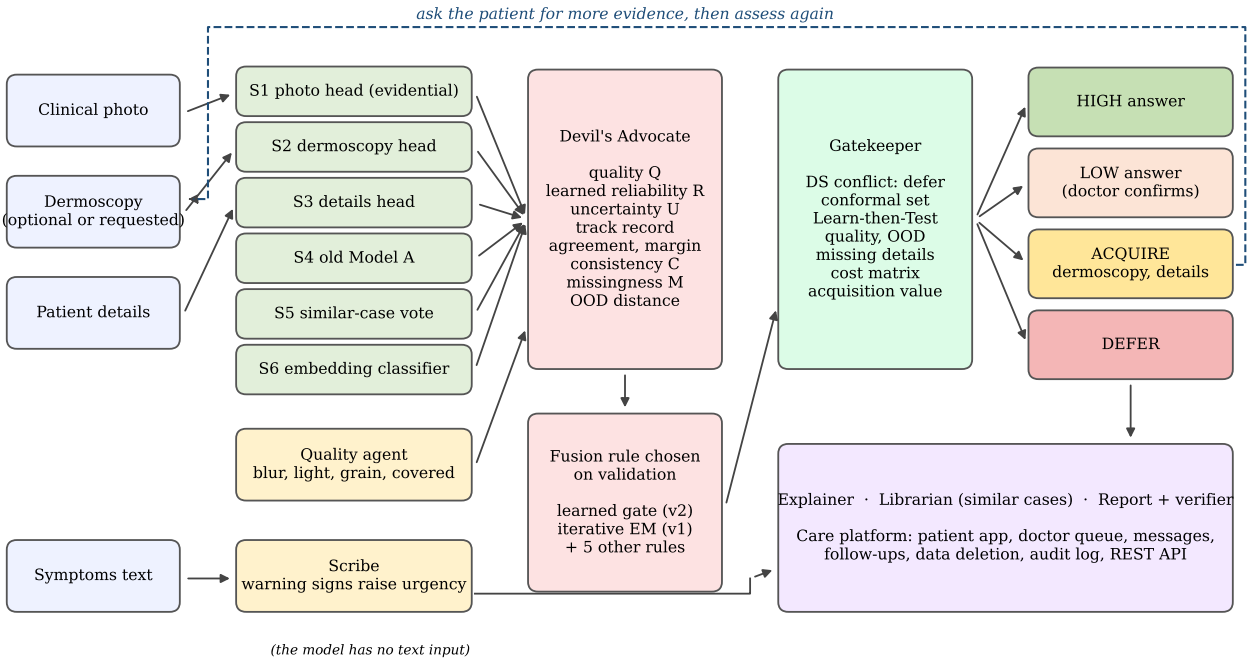


Fig. 1. Derma-Guard. Six specialists are profiled by the evidence manager ("Devil's Advocate") and combined by the fusion rule chosen on validation; the gatekeeper and a cost matrix choose one of four outcomes. The dashed loop is the request for more evidence, after which the case is assessed again.

B. The evidence manager

For specialist m on case x the evidence manager computes ten signals: quality Q_m (share of image statistics inside the training 5th–95th percentiles), a reliability R_m (logistic regression fitted on validation predicting whether m is right), normalised entropy U_m, confidence, margin, agreement with the other specialists, the measured track record $\pi_m(\hat{y}_0) = \Pr(m \text{ correct} \mid y = \hat{y}_0)$ at the provisional diagnosis, the team's Jensen–Shannon consistency [32], missingness, and the Mahalanobis out-of-distribution distance [18], [19]. The earlier prototype multiplied hand-set versions, $E_m = Q_m R_m U_m C S M_m$, and normalised them into weights; because consistency C and sufficiency S are equal for all specialists of a case, they cancelled, and only three signals ever moved the weights.

C. Fusion rules and how one is chosen

Iterative diagnosis-adaptive fusion (v1) keeps the prototype's EM-style loop [33]: weights $w_m \propto (Q_m R_m)^a$ give a provisional diagnosis $\hat{y}$; then up to three times $w_m = \mathrm{softmax}_m\left(a \log Q_m R_m + b \log \pi_m(\hat{y}) + g \log \kappa_m\right)$, with $\kappa_m = \rho$ for specialists disagreeing with $\hat{y}$, and the diagnosis is updated.

The learned gate (v2) lets every signal act. With the standardised signal vector $\phi_m(x)$,

$$w_m(x) = \underset{m \in \text{present}}{\mathrm{softmax}}\left(\beta_m + \theta_m^\top \phi_m(x)\right), \tag{2}$$

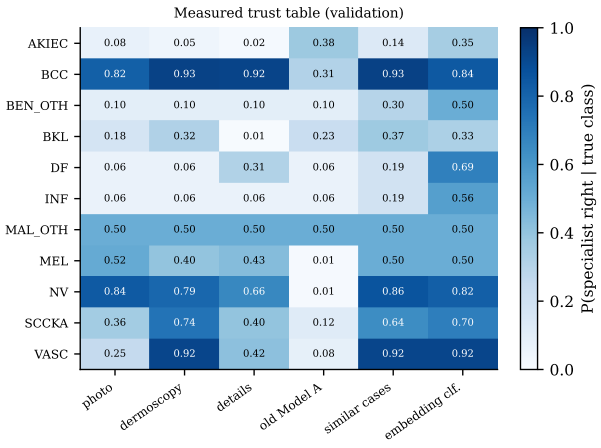


Fig. 2. The measured trust table $\pi_m(c)$: on validation lesions of each true class, how often each specialist alone names it. The prototype typed this table in by hand; for melanoma it assumed 0.96 for the photograph, where 0.52 is measured.

and the opinions are pooled either as a mixture $\sum_m w_m p_m$ or as a product of experts $\propto \prod_m p_m^{w_m}$, followed by temperature scaling. θ, β and the temperature minimise validation log-loss with an ℓ_2 penalty.

Seven rules are compared: simple average, the original formula with learned reliability, the single specialist with the highest R_m, the iterative fusion, stacking, and the gate

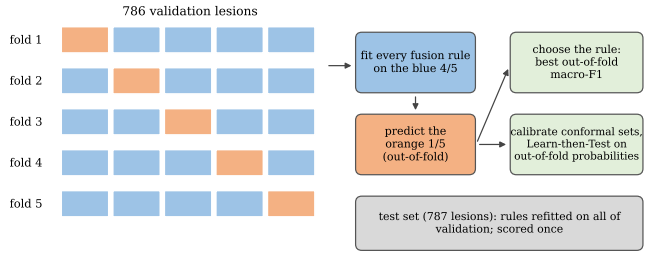


Fig. 3. Cross-fitting. Every fusion rule is fitted on four validation folds and predicts the fifth; the out-of-fold predictions both choose the rule and calibrate the safety guarantees, so neither step judges a model on its own training data.

with both poolings. Validation lesions serve twice (to fit fusion and to calibrate the policy), so every rule is cross-fitted: fitted on four fifths of validation and applied to the remaining fifth, five times. The out-of-fold probabilities are used both to select the rule (best macro-F1, fixed before any test result was seen) and to calibrate the policy. We also refit the selected rule on all 31 (stage A) and 63 (stage B) specialist subsets, and report an oracle that picks, per case, the specialist giving the true class most mass. Fig. 3 summarises the protocol.

D. The decision policy

Stage A uses the photograph and metadata; stage B adds dermoscopy. The stated diagnosis is the most probable class; costs are 0 for a correct answer, 1 within the benign or malignant group, $c_{\text{miss}} = 5$ for a cancer called benign and 2 for the reverse, with 0.5 to acquire dermoscopy and 1.5 for a clinician review. Gates are: Dempster–Shafer conflict above its validation 90th percentile (Defer); a Mondrian least-ambiguous conformal set over {benign, malignant} ($\alpha = 0.1$) [27], [28] that still contains "malignant" when the answer would be benign; a Learn-then-Test threshold [29] on $1-u$ such that $\Pr(\text{missed-cancer rate} \leq 0.05) \geq 0.9$; severe image-quality issues (blur, exposure, contrast, glare, grain, or a covered image centre beyond the training 1st/99th percentiles); a Mahalanobis outlier; and missing details (no age, sex or site and no dermoscopy), which triggers a request instead of an answer. If no gate fires and answering is cheapest, the answer is High. Otherwise the system requests dermoscopy when the expected cost after acquisition, estimated by a gradient-boosted regressor fitted on validation lesions whose dermoscopy is known, plus 0.5, is below the cost of review; else it gives a Low answer for a clinician to confirm. Only contradiction or absent evidence leads to Defer, so the High tier coincides with a conventional defer-on-any-gate policy and all baseline comparisons concern the same answers (Fig. 4). An earlier version deferred on every failed check, which sent every patient without a confident answer to a clinician with nothing at all; the Low tier keeps those patients informed while still requiring confirmation.

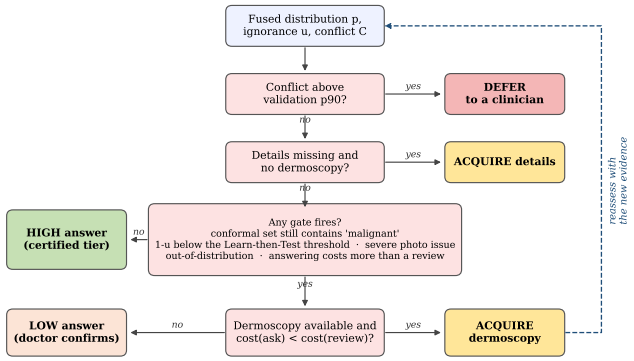


Fig. 4. How the policy chooses an outcome. Requests for evidence loop back: once the dermoscopic image or the missing details arrive, the case is assessed again.

E. Agents and platform

Around the policy, a quality agent advises on retakes, a scribe raises triage urgency when the patient's own words mention warning signs (with negation handling; it never changes the diagnosis), a librarian shows similar confirmed training lesions [34], an explainer states who said what, why dissenting specialists disagree and where the weight went, and a report writer may use a local language model whose output is discarded if a verifier finds any diagnosis or percentage not in the evidence. A care platform built on SQLite routes each outcome: a confident answer schedules a follow-up, a request moves the case to "awaiting patient" until the evidence arrives, and Low answers and deferrals enter a clinician queue ordered by kind and urgency. A follow-up photograph whose probability of malignancy rises by 0.15 or more returns the case as urgent. Every assessment stores the policy version that produced it, every event goes to an append-only audit log, and patients can delete a case or their whole record, including uploaded photographs. A Streamlit application exposes patient, clinician, stress-testing and research views, and a "compare systems" view runs any held-out lesion through every specialist, every fusion rule and every baseline side by side; a FastAPI service exposes the same workflow. The code base has 230 automated tests.

For one held-out melanoma in our test set, for instance, the explainer reports that the photograph head named melanoma but was only 10% sure, that the legacy model said benign keratosis because it "was trained on a different dataset and cannot name five of the eleven diagnoses" and was right on only 1% of validation melanomas, and that most of the weight went to the metadata head, the embedding classifier and the photograph head, which agreed with the team and had high learned reliability for this case. The system then asked for the dermoscopic image. Every number in such an explanation is read from the evidence profile; the language model, when used, may

TABLE I
Specialists alone and the main fusion rules on the test split (stage B: with dermoscopy). Val: out-of-fold validation macro-F1 used for selection.

	Val	Acc.	F1	Bal.	AUROC
Specialists (stage B)					
S1 photograph head	–	0.595	0.264	0.266	0.787
S2 dermoscopy head	–	0.705	0.400	0.411	0.854
S3 metadata head	–	0.600	0.246	0.252	0.749
S4 legacy model	–	0.219	0.077	0.096	0.553
S5 nearest-neighbour vote	–	0.719	0.431	0.402	0.866
S6 embedding classifier	–	0.700	0.440	0.447	0.875
Fusion, stage A (photograph + metadata)					
Dempster–Shafer, heads	0.274	0.644	0.286	0.280	0.799
Simple average	0.344	0.637	0.343	0.338	0.824
Iterative EM (v1)	0.374	0.670	0.383	0.375	0.832
Stacking	0.272	0.644	0.308	0.296	0.822
Learned gate (v2)*	0.412	0.657	0.377	0.360	0.838
Fusion, stage B (+ dermoscopy)					
Dempster–Shafer, heads	0.361	0.690	0.376	0.378	0.856
Simple average	0.457	0.714	0.422	0.400	0.870
Iterative EM (v1)	0.492	0.717	0.459	0.444	0.862
Stacking	0.413	0.720	0.372	0.361	0.853
Learned gate (v2)*	0.517	0.731	0.438	0.418	0.885
Oracle best specialist	–	0.840	0.594	–	–

*selected (product pooling). All seven rules: Fig. 5.

TABLE II
Decision policy and baselines on the same probabilities (test, $c_{\text{miss}} = 5$). Abstaining baselines are matched to Derma-Guard's coverage.

Policy	Ans.	Derm.	Acc.	Missed	Cost
Always answer, photograph	100%	0%	0.657	7.2%	0.759
Always acquire, answer	100%	100%	0.731	6.4%	1.124
Confidence threshold (B3)	57.1%	0%	0.815	3.8%	0.881
Learned rejector (L2D)	57.1%	0%	0.764	4.9%	–
Entropy-chosen acquisition	100%	29.5%	0.700	5.6%	0.790
Derma-Guard, High tier	57.1%	29.5%	0.771	3.1%	1.047
Derma-Guard outcomes: High 57.1%, Low 32.0% (top-3 85.7%), Defer 10.9%					

only reword it.

V. Results

A. Specialists and fusion

Table I and Fig. 5 summarise the specialists and the fusion rules. Dermoscopy carries real information: the dermoscopy head beats the photograph head on every metric, and adding it lowers the network's mean ignorance from 0.676 to 0.439. The legacy model transfers poorly (21.9% accuracy). The two case-based specialists are the strongest single sources, and the class-balanced classifier S6 has the best balanced accuracy.

Every way of weighting the team that uses the evidence beats the evidential network alone: without dermoscopy, macro-F1 rises from 0.286 to 0.377 for the selected gate, and with dermoscopy the gate has the best accuracy (0.731) and AUROC (0.885). The gate won the cross-fitted validation comparison in both stages; on test the iterative fusion has a slightly higher macro-F1, a reminder that macro-F1 over classes with fewer than ten test lesions is noisy. We kept the pre-declared choice. Stacking, the textbook learned combiner, is the weakest learned rule: 66 free inputs and 786 validation lesions are not enough. Adding S6 raised the selected rule's stage-A macro-F1 from 0.332 to 0.377, and the subset study found no subset that beat the full team by more than 0.033 on validation. The oracle's 0.594 shows that per-case selection, not the specialists, is the bottleneck. On average the gate gives the largest weights to the metadata head (0.35–0.41) and S6 (0.21–0.22), moves weight to dermoscopy when it is present, and gives the legacy model 0.07. Isotonic calibration makes the displayed confidence honest: 0.757 shown on average against 0.731 actual accuracy with dermoscopy, 0.660 against 0.657 without (Fig. 7).

Per class (Fig. 6), the team helps most where the evidential network was weakest: benign keratoses (F1 0.16 to 0.51), actinic keratoses (0.24 to 0.47) and the small "benign other" and inflammatory classes, which the network never predicted. It is worse on vascular lesions (0.88 to 0.36 on eight test lesions), where the dermoscopy head alone was the best judge, and the single malignant-other lesion is never found.

B. Decisions

Derma-Guard answers 89.1% of patients (Table II). Its 57.1% high-confidence answers are 77.1% accurate (95% CI 73.0–80.8%) and miss 3.1% of the cancers they answer (1.6–4.8%); Learn-then-Test certified thresholds at both stages ($\hat{\lambda} = 0.218, 0.276$). Low-confidence answers are right only 49.6% of the time, which is why they always go to a clinician, but the truth is in their top three for 85.7%. Dermoscopy was requested for 29.5% of patients.

At the same coverage, a learned rejector trained to predict the model's own errors [22] misses 4.9% of cancers, significantly more than Derma-Guard (+1.8 points, CI +0.2 to +3.5, one-sided $p = 0.011$). Plain confidence thresholding misses 3.8%; the difference (+0.7 points, CI −0.7 to +2.0, $p = 0.18$) is not significant, and thresholding is more accurate (81.5% vs. 77.1%) because it answers only the photograph's easiest cases. Always answering is cheapest in mean cost, since every clinician review is charged 1.5. Fig. 8 places these points on the full risk–coverage curves: no coverage level of the threshold reaches the policy's miss rate at 57% coverage.

Is the gap to thresholding real? We ran the identical policy on four fusion back-ends (Fig. 9). Its miss rate barely moved, 3.0–3.5%. The threshold's moved almost threefold, from 3.5% (the five-specialist gate) to 9.3% (the five-specialist iterative fusion). A confidence threshold is exactly as safe as the ranking it receives, and nobody deploying it knows in advance which kind of ranking they have; the calibrated policy delivers the same certified rate on all four. The same comparison contains a result against us: the learned acquisition estimator chose whom to send

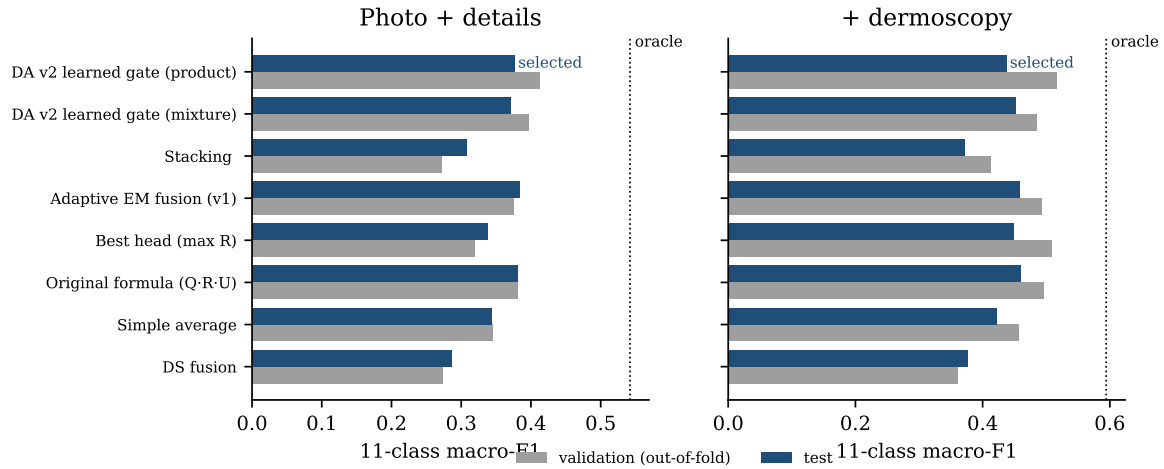


Fig. 5. All seven combination rules: out-of-fold validation macro-F1 (grey, used to select) and test macro-F1 (blue). The dotted line is the oracle that always picks the best specialist.

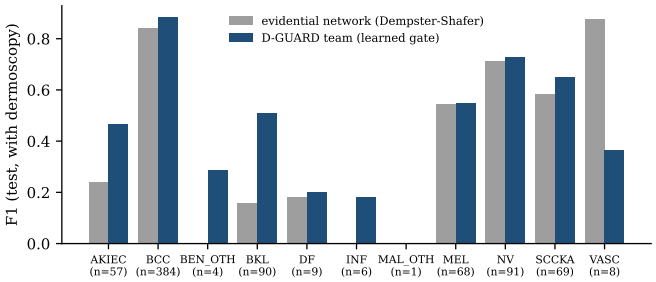


Fig. 6. Per-class F1 on the test set with dermoscopy: the evidential network alone and the full team. n is the number of test lesions.

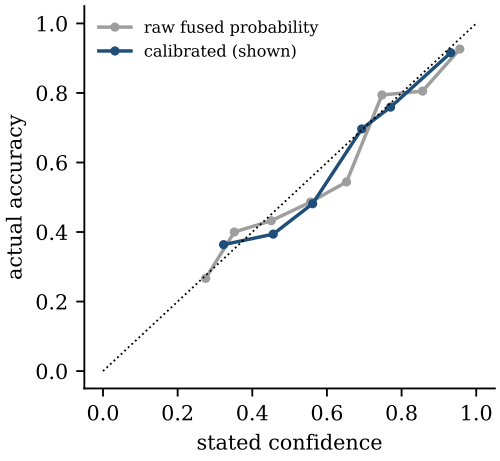


Fig. 7. Reliability diagram on the test set: the calibrated confidence that the application displays tracks the observed accuracy.

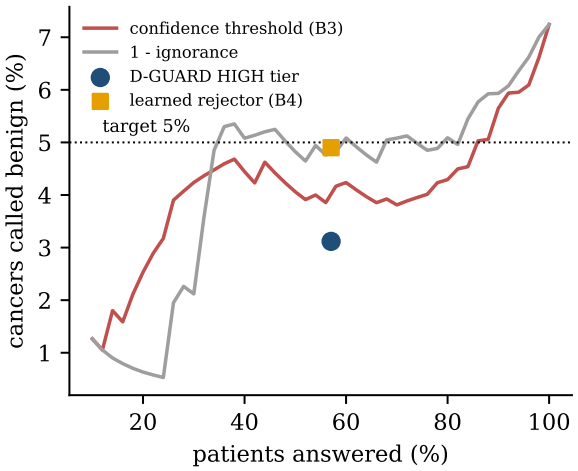


Fig. 8. Share of answered cancers called benign against the share of patients answered, for abstention by confidence and by $1-u$; the points are Derma-Guard's High tier and the learned rejector at the same coverage.

for dermoscopy far better than entropy on the iterative fusion (AE +0.30 vs. −0.04) but worse on the final gate (+0.07 vs. +0.26), and ignorance ranked errors better than softmax on the former (AUGRC 0.0874 vs. 0.0918) but not the latter (0.0820 vs. 0.0757). Once the fused probabilities are good, simple uncertainty is a good enough guide.

Raising c_{miss} from 2 to 10 lowers the High share from 61.6% to 53.4% and the miss rate from 4.1% to 1.9% (thresholding: 3.8% at the same coverage), while about 89% of patients still get an answer (Fig. 10). The cost of a missed cancer is therefore a dial a deploying clinic can set, not a hidden constant.

Fairness is a weakness (Fig. 11): for the 67 test lesions of skin tone 2 the High miss rate is 11.5%, against 2.2% for tone 3 ($n = 489$); tone 1 has only 14 lesions. MILK10k is dominated by tone 3, and a calibration fitted on the whole validation set guarantees an average, not every group; group-wise calibration needs more lighter-tone data than the dataset contains.

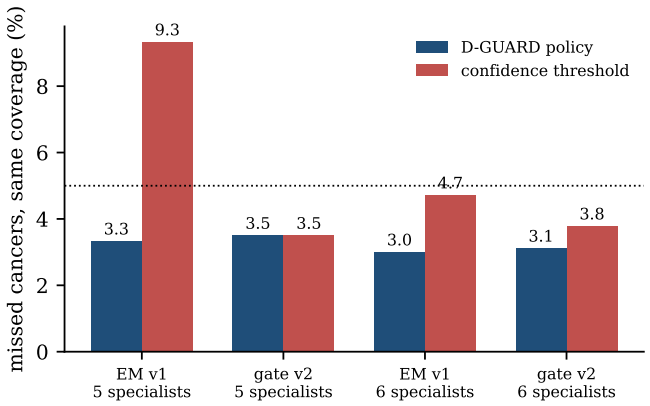


Fig. 9. The identical policy over four fusion back-ends (iterative EM or learned gate, with five or six specialists). The policy's missed-cancer rate stays at 3.0–3.5%; confidence thresholding at the same coverage ranges from 3.5% to 9.3%.

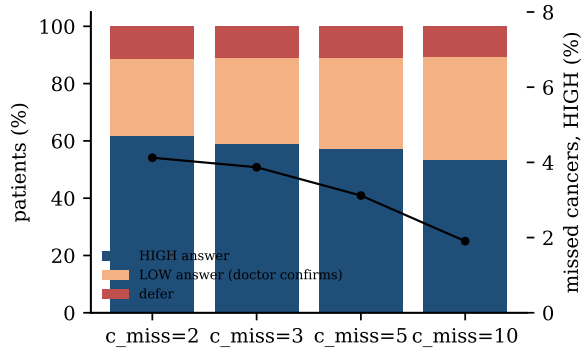


Fig. 10. Outcome shares (bars) and the missed-cancer rate of High answers (line) as the cost of a missed cancer increases.

C. Robustness: DERMA-STRESS

We damaged the photograph of 300 test lesions six ways without retraining (Fig. 12). Evidential ignorance does not catch what matters. Gaussian blur lowers mean ignorance (0.679 to 0.605) and raises the high-confidence rate (46% to 50%); under-exposed and occluded photographs keep 22% and 16% high-confidence answers that miss 12.3% and 14.3% of cancers. Explicit checks do catch it: the quality agent flags every blurred, dark, noisy and occluded photograph (the last two through the grain and covered-centre checks), and every case without metadata becomes a request for it. With these checks no damaged input received a high-confidence answer; JPEG artefacts kept 22% at 1.5% missed. The price is that 8% of clean photographs are also flagged.

What each component adds. Reading the experiments as an ablation: weighting the specialists by their evidence instead of fusing only the evidential heads adds 0.091 photograph-only macro-F1 (0.286 to 0.377); the sixth specialist alone accounts for 0.045 of it; cross-fitting removed an optimistic three points of High coverage that

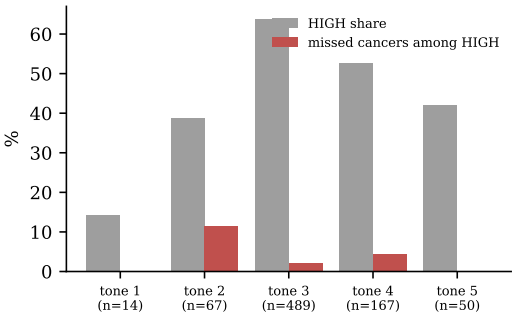


Fig. 11. High share and missed cancers among High answers by MILK10k skin-tone class.

TABLE III
The MILK10k system on two other datasets, unchanged. Threshold: confidence thresholding at the same coverage.

	MILK10k	PAD-UFES-20	Derm7pt
Dermoscopy available	yes	no	on request
Malignant share	72%	79%	29%
Photos failing quality check	8%	28%	29%
Dermoscopy requested	30%	28%	39%
Accuracy, all cases	0.731	0.367	0.479
Malignant AUROC	0.885	0.840	0.738
High share	57.1%	11.4%	13.0%
Accuracy of High answers	0.771	0.517	0.708
Missed cancers, High	3.1%	0.0%	3.8%
Threshold: missed / accuracy	3.8% / 0.815	0.0% / 0.574	5.4% / 0.846

in-sample fitting had produced; the calibrated policy, rather than any particular fusion, is what keeps the miss rate at 3.0–3.5%; and the explicit image and missing-details checks, not evidential ignorance, are what remove confident answers on damaged input.

D. External tests

The calibration guarantees assume that future patients resemble calibration patients. Table III shows what happens when they do not. On both external datasets ignorance, quality flags and deferrals rise and the high-confidence share falls from 57% to 11–13%. Among the confident answers that remain the safety target holds: no cancer was missed on PAD-UFES-20 and 3.8% on Derm7pt, below thresholding's 5.4%. On Derm7pt, where dermoscopy exists, the system asked for it for 39% of lesions and overall accuracy rose from 0.402 with photographs alone to 0.479. The eleven-class diagnosis does not transfer: half of PAD-UFES-20's confident answers are wrong, melanoma recall on Derm7pt is 31%, and thresholding is more accurate on both datasets (Fig. 13). Nothing was retrained on either dataset; they are external tests, deliberately kept out of training so that they measure what a deployment would meet.

E. The earlier prototype, re-evaluated

The prototype trained seven PAD-UFES-20 models (image, metadata, free text and their combinations) and

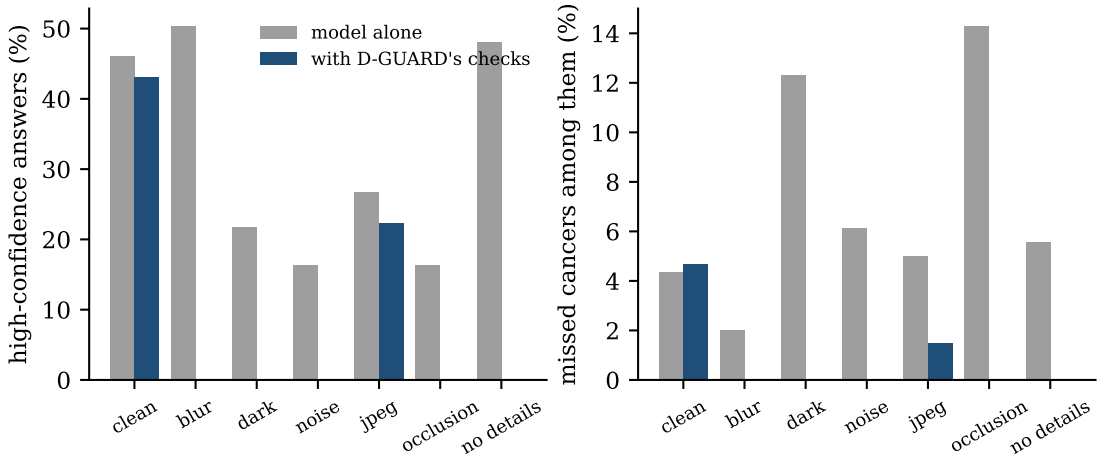


Fig. 12. DERMA-STRESS on 300 test lesions (photograph + metadata): share of high-confidence answers and missed cancers among them, for the model alone (grey) and with the pipeline's quality and missing-details checks (blue).

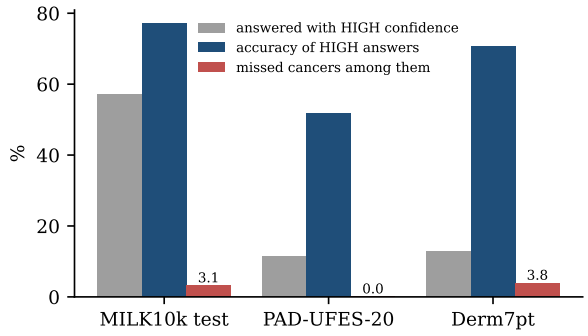


Fig. 13. Share of High answers, their accuracy and the cancers missed among them, on MILK10k and the two external datasets.

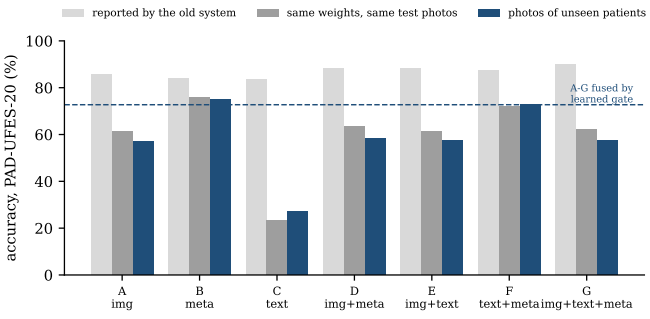


Fig. 14. The earlier prototype's seven PAD-UFES-20 models: accuracy as reported, re-scored with the same weights on the same test photographs, and on photographs of patients never seen in training. Dashed: the seven fused by Derma-Guard's learned gate (cross-validated).

reported 84–90% accuracy. Its split was random by photograph, so 43% of test photographs came from a lesion and 62% from a patient already in training, and its evaluation function swapped probabilities between the true class and its closest rival until the number of correct predictions

TABLE IV
The prototype's models on PAD-UFES-20: reported accuracy and honest re-scoring of the same saved weights.

	Inputs	Reported	Same test	Unseen patients
A	image	85.5%	61.4%	56.8%
B	metadata	84.1%	75.9%	75.0%
C	free text	83.5%	23.5%	27.3%
D	image + metadata	88.4%	63.5%	58.3%
E	image + text	88.1%	61.4%	57.6%
F	text + metadata	87.2%	72.2%	72.7%
G	all three	89.9%	62.0%	57.6%
A–G fused by the learned gate (unseen patients)				72.7%
A–G simple average (unseen patients)				59.8%

met a hard-coded target per model. Loading the saved weights and cached features unchanged and scoring them honestly (Fig. 14): on the 132 test photographs of unseen patients, the image model reaches 56.8%, the metadata model 75.0%, the text model (whose "narratives" were generated from the metadata) 27.3%, and the three-modality model 57.6%, worse than metadata alone. Fusing the seven with our learned gate, cross-validated on those 132 photographs, gives 72.7% against 59.8% for averaging them; the gate recovers almost all of the best model's accuracy without being told which model that is.

Finally, the PAD-UFES-20 questionnaire lets us test the scribe's triage rule (Fig. 15): of 1,641 lesions, those with any of its warning signs were malignant (BCC, SCC or melanoma) in 48.7% of cases against 12.4% without (relative risk 3.9, 95% CI 2.8–5.5), and every single sign had a relative risk between 1.65 and 3.0. A raised lesion, which the scribe did not originally look for, carried a relative risk of 2.8 and was added as a result. These are clinician-collected questionnaire answers rather than free text, so the analysis supports the rule's direction, not its exact wording.

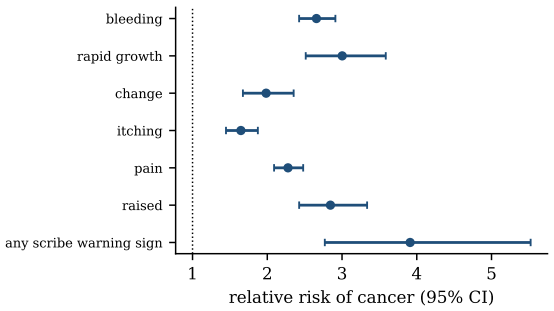


Fig. 15. Relative risk of cancer with each warning sign on the PAD-UFES-20 questionnaire (95% CI). All exceed one.

VI. Discussion

What the earlier design got right. Consulting several specialists, questioning their evidence, and trusting them according to the suspected diagnosis all survived honest evaluation; every evidence-aware rule beat both the evidential network alone and plain averaging, and the measured trust table confirms clinical intuition (dermoscopy is reliable for squamous lesions, the photograph for melanoma). What it got wrong was its evidence.

What did not work. Stating the minimum-cost diagnosis (classic Chow [20]) labelled 98.5% of lesions malignant on this 72%-malignant population; adaptive prediction sets for the safety gate collapsed to the full set and deferred every patient; post-hoc logit adjustment raised validation macro-F1 from 0.36 to 0.44 but lowered test macro-F1 from 0.376 to 0.359; fitting the fusion in-sample on the calibration lesions inflated the high-confidence share by three points before cross-fitting; and classes with fewer than ten test lesions are almost never predicted.

Where the gain comes from. The oracle that picks the best specialist per case reaches 0.594 macro-F1, against 0.438 for the learned gate. The specialists already contain much of the right answer; deciding whom to believe is the hard part. A gate trained on more validation data, or distilled from the oracle, is the most direct way forward, and a dermatology foundation backbone [5] would lift every specialist.

Lessons about evaluation. Split by the unit you will meet in practice; never post-process predictions with the label; choose on validation and keep the choice when the test disagrees; and compare against simple baselines on the same probabilities, and on more than one model. The most useful finding of this work, that thresholding's safety is borrowed from the fusion beneath it, appears only under that last rule.

Limitations. All results come from one training seed of an ImageNet ResNet50; 786 validation lesions do a great deal of fitting; MILK10k is a referral population without patient or centre identifiers; the costs are declared, not derived; the guarantees assume exchangeability; the image checks are heuristics; lighter skin tones are under-served; and the system has not been evaluated in a clinic.

VII. Conclusion

Derma-Guard turns a team of imperfect specialists into calibrated clinical actions: it answers nine in ten patients, more than half with a confident answer that misses about one cancer in thirty, asks for dermoscopy, details or a better photograph when they are needed, and refers only when the evidence contradicts itself. Learning how much to trust each specialist from all of its evidence improves photograph-only diagnosis by a third. The calibrated policy is safer than a learned rejector and, unlike confidence thresholding, stays safe whichever fusion it is given. We hope the most transferable lesson is methodological: report the simple baseline on the same probabilities, on more than one model, and keep every number reproducible.

Reproducibility

Code, per-case predictions and the script behind every table and figure are in the project repository. Model weights and images are not redistributed; MILK10k is available from the ISIC Archive under CC-BY-NC.

References

[1] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," Nature, vol. 542, pp. 115–118, 2017.

[2] N. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, H. Kittler, and A. Halpern, "Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (ISIC)," arXiv preprint arXiv:1902.03368, 2019.

[3] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," Scientific Data, vol. 5, p. 180161, 2018.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.

[5] S. Yan et al., "A multimodal vision foundation model for clinical dermatology," Nature Medicine, 2025, arXiv:2410.15038.

[6] A. G. C. Pacheco and R. A. Krohling, "An attention-based mechanism to combine images and metadata in deep learning models applied to skin cancer classification," IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 9, pp. 3554–3563, 2021.

[7] J. Kawahara, S. Daneshvar, G. Argenziano, and G. Hamarneh, "Seven-point checklist and skin lesion classification using multi-task multimodal neural nets," IEEE Journal of Biomedical and Health Informatics, vol. 23, no. 2, pp. 538–546, 2019.

[8] A. G. C. Pacheco et al., "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," Data in Brief, vol. 32, p. 106221, 2020.

[9] MILK study team, "MILK10k: Multimodal imaging of lesions with knowledge," International Skin Imaging Collaboration (ISIC) Archive, 2025, cC-BY-NC.

[10] D. Mehta, C. Primiero, B. Betz-Stablein, T. D. Nguyen, Y. Gal, A. Bowling, M. Haskett, M. Sashindranath, P. Bonnington, V. Mar, H. P. Soyer, and Z. Ge, "Multi-task AI models in dermatology," Journal of the European Academy of Dermatology and Venereology, vol. 39, no. 12, pp. 2121–2133, 2025.

[11] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in Advances in Neural Information Processing Systems (NeurIPS), 2018, pp. 3179–3189, arXiv:1806.01768.
[12] A. Jøsang, Subjective Logic: A Formalism for Reasoning Under Uncertainty. Springer, 2016.
[13] A. P. Dempster, "Upper and lower probabilities induced by a multivalued mapping," The Annals of Mathematical Statistics, vol. 38, no. 2, pp. 325–339, 1967.
[14] G. Shafer, A Mathematical Theory of Evidence. Princeton University Press, 1976.
[15] Z. Han, C. Zhang, H. Fu, and J. T. Zhou, "Trusted multi-view classification," in International Conference on Learning Representations (ICLR), 2021, arXiv:2102.02051.
[16] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in International Conference on Machine Learning (ICML), 2017.
[17] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2002.
[18] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in Advances in Neural Information Processing Systems (NeurIPS), 2018.
[19] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," Journal of Multivariate Analysis, vol. 88, no. 2, pp. 365–411, 2004.
[20] C. K. Chow, "On optimum recognition error and reject tradeoff," IEEE Transactions on Information Theory, vol. 16, no. 1, pp. 41–46, 1970.
[21] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in Advances in Neural Information Processing Systems (NeurIPS), 2017.
[22] H. Mozannar and D. Sontag, "Consistent estimators for learning to defer to an expert," in International Conference on Machine Learning (ICML), 2020.
[23] J. Traub, T. J. Bungert, C. T. Lüth, M. Baumgartner, K. H. Maier-Hein, L. Maier-Hein, and P. F. Jäger, "Overcoming common flaws in the evaluation of selective classification systems," in Advances in Neural Information Processing Systems (NeurIPS), 2024, spotlight. arXiv:2407.01032.
[24] V. Vovk, A. Gammerman, and G. Shafer, Algorithmic Learning in a Random World. Springer, 2005.
[25] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," arXiv preprint arXiv:2107.07511, 2021.
[26] Y. Romano, M. Sesia, and E. J. Candès, "Classification with valid and adaptive coverage," in Advances in Neural Information Processing Systems (NeurIPS), 2020.
[27] M. Sadinle, J. Lei, and L. Wasserman, "Least ambiguous set-valued classifiers with bounded error levels," Journal of the American Statistical Association, vol. 114, no. 525, pp. 223–234, 2019.
[28] C. Lu, A. Lemay, K. Chang, K. Höbel, and J. Kalpathy-Cramer, "Fair conformal predictors for applications in medical imaging," in Proceedings of the AAAI Conference on Artificial Intelligence, 2022, arXiv:2109.04392.
[29] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," The Annals of Applied Statistics, 2025, arXiv:2110.01052.
[30] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," arXiv preprint arXiv:2208.02814, 2022.
[31] T. M. Cover and P. E. Hart, "Nearest neighbor pattern classification," IEEE Transactions on Information Theory, vol. 13, no. 1, pp. 21–27, 1967.
[32] J. Lin, "Divergence measures based on the Shannon entropy," IEEE Transactions on Information Theory, vol. 37, no. 1, pp. 145–151, 1991.
[33] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," Journal of the Royal Statistical Society: Series B, vol. 39, no. 1, pp. 1–38, 1977.
[34] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems (NeurIPS), 2020.